FLEET TEAM DIGEST

737MAX-FTD-27-20001

Issue Title : Stabilizer Trim Motor (STM) Low Time Failures

Airplane Model	ATA	Minor Model(s)
737MAX	2741-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
04/27/2021	06/02/2026	06/02/2026

Estimated Completion Date	Next Update Date

Revision Description

Revised Interim Action, Final Action and Milestones. This update closes the issue and FTD.

Description

This FTD provides information regarding design changes and reliability of the Stabilizer Trim Motor (STM) P/N 6355D0001-01.

Since introduction of the STM, redesigned for use on the 737 MAX aircraft, Boeing has experienced a number of early STM failures in Production. These early failures have occurred in the Renton Factory, Renton flight line and Boeing Field. Operators have also reported ocasional low time removals.

Background

Eaton identified a potential Printed Wire Assembly (PWA)/component tolerance concern. Eaton revised the Printed Wiring Board (PWB) to ensure proper component fit. The Insulated Gate Bipolar Transistor (IGBT) and heatsink hole tolerance was increased to ensure proper assembly. Eaton also implemented component screening of the IGBT. The control PWA, part number was changed from 6355-1230-05 to 6355-1230-06. STMs with the revised Control PWA are identified as Mod 1.Eaton later made two additional mod level changes to accommodate changes in the “commercial off the shelf” IGBT that was impacting production yield and infrequent production failures. The changes to the motor drive circuit includes the replacement of nine resistors and three capacitors. These changes create the -07/-08 control PWA configurations. Retrofit of a -05 results in a -07 Control PWA configuration. Retrofit of a -06 results in a -08. STMs that have a -07 are identified as Mod 2. STMs using the -08 are identified as Mod 3. Service Bulletin 6355D0001-27-1 was released to upgrade STMs with -05 Control PWA (Mod 0) and -06 Mod 1 to Mod 2 and Mod 3, respectively. In the fourth quarter of 2020, an Operator reported several low time failures of the Stabilizer Trim Motor (STM) Part Number 6355C0001-02 Mod 2. At Eaton, analysis of the returned STMs showed failure was due to Electrical Over Stress (EOS) of the Control or Filter PWAs.

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As a result of this trend Boeing and Eaton conducted a Root Cause / Corrective Action (RCCA) investigation.

The RCCA identified the primary root cause as improved switching performance in the IGBT coupled with a software issue. The IGBT components, currently available, have a significantly higher current capacity and faster switching times which has made a previously unnoticed software behavior cause failure due to EOS. The current software, allows the upper IGBT in one phase to turn on while the lower IGBT in the same phase is on (or vice versa) resulting in a high current “shoot through" event. Additionally, Boeing and Eaton are making a change to the control PWA that will add a ferrite bead and polyimide film tube to the IGBT’s gate lead. This modification will reduce high frequency noise, identified during the RCCA investigation, affecting ATP pass rates. This change will introduce a Mod 4/5 configuration. This change alone is not expected to improve STM reliability.

Failure of the STM is detected by the flight crew by no response to input from the control wheel mounted main electric trim switches or STAB OUT OF TRIM light illumination if the auto pilot is in use.

Crew should follow the Quick Reference handbook (QRH) 9.26 Stabilizer Trim Inoperative.

Status

Boeing and Eaton have completed the hardware and software redesign activities (Requirements Analysis, Preliminary Design and Critical/Final Design).

Eaton completed Qualification testing.

Boeing has completed ground and flight test with no technical issue identified.

Eaton released Service Bulletin (SB) 6355D0001-27-02 to upgrade 6355D0001-01 to -02 on 28 June 2024.

Interim Action

None, final action has been released.

Final Action

Implement Eaton Service Bulletin 6355D0001-27-02 to upgrade part number 6355D0001-01 Mod 0, 1, 4 or 5 to 6355D0001-02, on an attrition basis. Industry support information is contained in the service bulletin. Operators should contact Eaton to obtain cost and schedule information, if planning a fleet-wide campaign.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

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Root Cause Established: Complete - June 2021

Solution Selected: Complete - September 2021

Complete Final Design: Complete - October 2022

Complete Qualification Testing: Complete - February 2023

Complete Ground and Flight Testing: Complete - April 2023

Complete Certification: Complete April 2024

Component Service Bulletin Available: Complete 28 June 2024

Supplier Information

Eaton Aerospace LLC

Fuel and Motion Control Systems Division

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Grand Rapids, MI 49512-0872 USA

(616) 949-1090

GRCustomerService@Eaton.com

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Stabilizer Trim Motor	10-62233-8	Eaton Aerospace	6355D0001-01

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- CRC - Component Reliability Collaboration : Collaborative process to eliminate component removals in production and the during the first year of new airplane in service
- Significant Technical Flight Interruptions : Issues that cause in-flight events such as rejected takeoffs, air turnbacks, diversions, emergency landings, in-flight shutdowns, and abnormal stops

Attachments:

a) Stabilizer Trim Motor Mod level Identification

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